

Sep 26, 2025

Meeting Sep 26, 2025 at 09:04 PKT

Summary

Sualeh Ali outlined the requirements for Milestone 1, including a GitHub repository, architecture diagrams, setup guides, and a contribution markdown table, while also detailing the use of Docker, GitHub Actions, and MLOps tools like MLflow and Prometheus. Haris Naveed provided an in-depth overview of MLOps, distinguishing between prototypes and production systems, recommending essential MLOps tooling such as Kubernetes and MLflow, and discussing model optimization techniques like quantization, as well as the importance of trust and safety in AI, clarifying concepts like data and concept drift. Fizza Zehra, Ahmad Murtaza, and Agha Salik discussed their restaurant review project, with Sualeh Ali advising the integration of traditional ML techniques alongside sentiment analysis.

Details

- **Milestone 1 Requirements** Sualeh Ali outlined the requirements for Milestone 1, emphasizing the need for a GitHub repository with a clear README, an architecture diagram using tools like draw.io or Mermaid, a setup guide, and a contribution markdown table detailing roles and responsibilities. They also specified the use of Make for running the project and the implementation of a branch naming convention for version control.
- **Docker and Microservices** Sualeh Ali explained the requirement for a working Dockerfile, noting that each microservice should run in its own Dockerized container. They mentioned that bonus points would be awarded for projects utilizing a database and Prometheus.
- **GitHub Actions and Acceptance Tests** Sualeh Ali introduced GitHub Actions for CI/CD workflows, highlighting the need for pre-commit hooks to detect security

vulnerabilities and enforce checks before code merges. They also detailed the requirement for acceptance tests to ensure the application meets benchmarks before deployment, ideally running in a canary environment.

- **MLOps Tools and Monitoring** Sualeh Ali discussed the importance of MLOps tools like MLflow for experiment tracking and Prometheus and Grafana for monitoring dashboards. They emphasized the need to expose metrics such as projected latency and number of requests, and also mentioned Evidently for machine learning drift detection.
- **Discussion on Monitoring Environments** Talha Shahid inquired about the application of Prometheus and Grafana for server utilization monitoring in pre-production versus production environments. Sualeh Ali clarified that for the class, a local environment is assumed, though in an ideal scenario, dev/staging and production servers would be segregated. Talha Shahid also asked if these tools work with cloud hosting, to which Sualeh Ali affirmed their compatibility with platforms like EC2 instances.
- **API Documentation and Security Checks** Sualeh Ali covered the necessity of API documentation, recommending Swagger or Fast API, and requested a cURL command in the README for testing service functionality. They also highlighted security best practices, including the use of tools like `pip audit` to identify vulnerabilities in outdated packages.
- **Guest Speaker Introduction and MLOps Overview** Sualeh Ali introduced Haris Naveed as an MLOps engineer at Motive, highlighting their expertise in the field and their background in ML engineering and data science. Haris Naveed then began their presentation by outlining the agenda, which included a broad overview of MLOps, a case study, concept and data drift, RAG-based systems, and trust and safety in machine learning.
- **Distinction Between Prototypes and Production Systems** Haris Naveed explained the difference between prototype development, which is flexible, and production systems, which are typically built to deliver shareholder value. They presented a typical production pipeline that incorporates automated model development, experiment tracking, and continuous monitoring to ensure model performance and facilitate retraining with new data.
- **Essential MLOps Tooling** Haris Naveed recommended several key MLOps tools, including Kubernetes for deployment and scaling services, MLflow for experiment tracking and model registries, and DataDog for comprehensive

monitoring and visualization dashboards. They also suggested the open-source combination of Prometheus and Grafana for monitoring and logging in personal projects, noting DataDog's paid nature.

- **Data Visualization and Deployment Tools** Haris Naveed discussed visualization tools like Datafana and Data Dog, highlighting Data Dog's integrated features for dashboards and alerts. He also introduced Argo CD as a continuous deployment platform that uses GitOps to sync configurations automatically.
- **AI Dash Cam and Edge Device Model Optimization** Haris Naveed explained Motive's AI-powered dash cam, which detects dangerous road and driver behaviors using edge-based vision models. He presented a case study on optimizing a food classification model for mobile devices, emphasizing the need to meet constraints on size, latency, and accuracy, and noting MobileNetV2 as a good starting point.
- **Model Quantization Techniques** Haris Naveed elaborated on post-training quantization as a method to reduce model size and improve inference speed by lowering precision, comparing it to going from a detailed map to a less detailed one. He differentiated between static and dynamic quantization, noting static quantization's speed advantage for convolutional neural networks. Haris Naveed demonstrated how Intel's Neural Compressor library can be used for quantization while maintaining accuracy.
- **Data Drift vs. Concept Drift** Haris Naveed clarified the distinction between data drift, which occurs when the underlying data distribution changes, and concept drift, where the fundamental relationship between variables changes. He provided a gym membership example to illustrate data drift, where an influx of non-athletes affected a weight prediction model trained on athletes. He also mentioned tools and methods like AETL's `ali` detector and maximum mean discrepancy for monitoring data drift in production, triggering alerts and retraining pipelines when necessary.
- **Trust and Safety in AI** Haris Naveed addressed the importance of trust and safety in AI, encompassing algorithmic fairness, transparency, interpretability, and data protection/privacy. He highlighted the challenge of biased models picking up proxies for sensitive attributes and the need for explainable AI. Haris Naveed emphasized that compliance with data privacy laws, such as GDPR, is crucial for companies operating in various markets.

- Organizations Promoting Responsible Technology** Haris Naveed recommended engagement with organizations focused on ethical technology development. He mentioned SIM, which builds tools for journalists and researchers to analyze narratives on social media, All Tech Is Human, which provides a community and job opportunities in trust and safety, and the Center for Humane Technology, founded by Tristan Harris, which advocates for human-centered technology design.
- Limitations of LLMs in High-Stakes Domains** Haris Naveed discussed the current limitations of Large Language Models (LLMs) in high-stakes domains like finance and healthcare, suggesting that human oversight will remain essential due to compliance requirements and the need for qualitative evaluation. Sualeh Ali supported this by sharing Karim's experience, where human intervention is required even for non-groundbreaking use cases to ensure accuracy and address legal implications.
- AI Risk and Opportunity** Haris Naveed discussed the risks and opportunities of AI, noting that while some places like Europe and California are seeing improvement in AI regulation due to data protection policies, the situation might worsen before it gets better. They explained that corporations prioritize revenue or publicity, necessitating government intervention to regulate them. Haris Naveed also mentioned that the "responsible AI" sector experienced a decline in research grants in 2023-2024, similar to a "winter" period, but predicted a resurgence of interest due to emerging problems or governmental changes.
- Impact of AI on Creative Fields and Professional Practices** Haris Naveed commented on the impact of AI on various fields, suggesting that creative fields like music, art, and video editing might see increased automation, but a human touch will still be highly valued, leading people to pay premium prices for human-created work. Haris Naveed also clarified that despite advancements in AI, mass layoffs due to automation have not occurred, and practical automation in areas like LinkedIn or self-driving cars still requires significant human oversight. When evaluating AI models, Haris Naveed advised linking model metrics to business KPIs to predict their impact on performance.
- Importance of Trust and Safety in AI** Sualeh Ali emphasized the critical importance of trust and safety in the AI domain, highlighting that this area is often overlooked despite offering significant opportunities for differentiation amidst intense competition in the AI field. Sualeh Ali noted that understanding the broader ecosystem around large language models (LLMs), including their

legal and social implications, is crucial for developing a diverse skill set in AI. Haris Naveed provided contact information for further queries, expressing gratitude for the audience's participation.

- **Restaurant Review Project Discussion** Fizza Zehra initiated a discussion about their project on restaurant reviews, and Sualeh Ali confirmed that it was an appropriate time to discuss it. Ahmad Murtaza and Agha Salik sought guidance on incorporating traditional machine learning (ML) aspects into their restaurant review tagging project, which focuses on sentiment analysis of Google reviews, similar to Yelp. Sualeh Ali suggested that the project could include traditional ML approaches like clustering or unsupervised methods for restaurant discovery, such as finding nearest neighbors or encoding user preferences and location. Sualeh Ali also advised that the project should incorporate sentiment analysis, which would be the LLM use case, alongside a numerical use case for tagging, and while starting with a static dataset is acceptable, programmatically updating the data would be ideal.

Suggested next steps

- ☐ Sualeh Ali will decide the ETA for milestone one, which is approximately the last week of October.
- ☐ Sualeh Ali will share the finalized directory, and cover GitHub actions, MLflow, Prometheus, Grafana, and model versioning, including DVC and Git LFS, in detail later.
- ☐ Sualeh Ali will finalize bonus points for those using GPUs and cover infrastructure as code which is terraform and K6 for load testing.
- ☐ Sualeh Ali will fix the issue with GitHub actions getting triggered.
- ☐ Sualeh Ali will upload the milestone one stuff by the end of this weekend.

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